

PERFORMANCE ASSESSMENT AND OPTIMIZATION OF A MICRO-HYDROPOWER PLANT: CASE OF KIGWENA IN BURUNDI

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Abstract:

A significant characteristic of the power sector in Burundi is a very low electrification rate (about 10 %), mainly dependent on hydro-power and with frequent power outages in dry seasons. Despite having potential for developing various renewable energy technologies, the Burundian Agency for Rural Electrification promoted usage of micro-hydropower plants. Hence, most of these plants are affected by stream-flow fluctuations especially during the dry seasons. This study aimed to identify a simple, reliable, viable and cost-effective hybrid power system to compensate the power supply intermittencies of the Kigwena micro-hydropower plant. The optimization of the best hybrid combination was evaluated using the microgrid software, HOMER Energy. The study concluded that the most suitable complementary energy resource was solar-PV and the utility grid for smoothing unexpected intermittencies. Economically, these renewable energies were found to be the most techno-economical viable option to upgrade the capacity of the hydropower with the lowest Levelized Cost of Energy of US\$ 0.0334 /kWh. This cost was within the range in comparison to the current electricity tariff in Burundi ranging from US\$ 0.023/ kWh to US\$ 0.072/kWh. The study would be a replicable model for other areas facing similar challenges.

Keywords: Renewable Energy, Micro-hydropower, Optimization, Hybrid power system

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I. INTRODUCTION

The access to affordable, reliable, sustainable and modern energy for all is proven to be a key for achieving other sustainable development [1]. The situation of universal energy access in Africa is alarming. The worst case is the Sub-Saharan region where 633 million people were estimated to experience a lack of electricity access, and another 792 million people have no choice but to cook with traditional biomass [2].

For the case of Burundi, the over-dependence on hydropower as the main energy source for its power supply has been hampering its economy. Its total power supply registered a reduction of 40 % due to a reduced volume of water in 2009 [3]. However, this country is endowed with many options for renewable energies (RE) such as hydropower [3] and solar [4]. As even more, there are not enough fossil fuel resources discovered yet, RE is a more important strategic component for diversification of its national energy supply [5]. Hence, efforts to facilitate a distributed generation system were facilitated by the Burundian Agency for Rural

Electrification (ABER), involved in construction of 5 small hydropower plants: Kigwena, Butezi, Kayongozi, Ryarusera and Nyabikere. However, these plants which rely on climatic conditions are facing power cut-offs, mainly during dry seasons [6]. Therefore, as electricity from the national grid was not sufficient to compensate for the fluctuations, the hybrid power system (HPS) was found to be a common solution for rural areas' electrification and this would improve the power supply reliability ([7], [8], [9], [10], [11]). The HPS optimization is used to minimize the system cost while the system reliability is maintained. However, most research in Burundi ([12]; [13]; [14]) has concentrated on the possible development of small hydro-power plants as a main option for reliable power supply. Therefore, this study sought to assess the best HPS to support the unreliable power supply of Kigwena MHP and increase its generation capacity to satisfy the demand. As this country is endowed with different RE resources [4], the study would pave the way for developing similar studies for reliable rural electrification in other locations of the country.

II. METHODOLOGY

The focus was on the evaluation of the power gap demand of the village, identify the performance Kigwena MHP and thus, upgrading the capacity of the plant to meet the demand. HOMER software was used to optimize available RE.

II.1. Performance assessment of the plant

The evaluation involved the values at different points of power components at the station (Fig. 1).

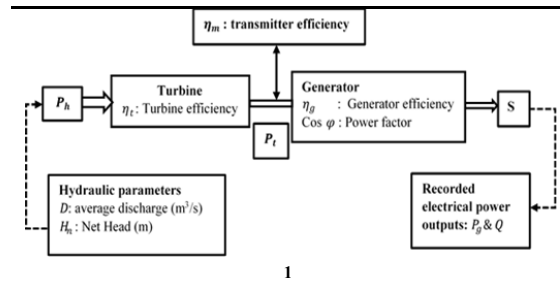


Fig. 1 The MHP performance evaluation

The power output of the generator is given by:

$$P_g = P_t \eta_g \eta_m \quad (1)$$

And the power at the turbine shaft is given by:

$$P_t = \eta_t P_h \quad (2)$$

The hydraulic power (available power at the turbine inlet) can be formulated as equation (3):

$$P_h = \rho g H_n D \quad (3)$$

The plant overall efficiency is given by equation (4):

$$\eta_0 = \frac{P_g}{P_h} = \eta_t \eta_g \eta_m \quad (4)$$

It is possible to determine the turbine power output using equations (5) and (6).

$$\text{Generator (kVA)} = \frac{(\text{Turbine output in kW}) * (\text{Generator efficiency})}{\text{Generator power factor}} \quad (5)$$

$$S = [P^2 + Q^2]^{1/2} \quad (6)$$

Where S [kVA] is the apparent power, P [kW] active power and Q [kVAR] reactive power.

II.2. Alternative energy resources in the area

i. Assessment solar energy potential

The Burundi Geographical Institute (IGEUBU) was not in possession of this data. A series of empirical equations (7) to (11) was used ([9], [15]).

$$H = H_0 \left(a + b \frac{n}{N} \right) \quad (7)$$

H is the monthly average daily radiation on horizontal surface (J/m²); H₀, the monthly average daily extra-terrestrial radiation on a horizontal surface (J/m²); N, monthly average of maximum possible daily hours of bright sunshine; n, monthly average daily hours of bright sunshine; a and b are empirical constants depending on location estimated to a = 0.30; b = 0.51 for this study [16].

H₀ is found using equation (8):

$$H_0 = \frac{24 * 3600 G_{sc}}{\pi} \left[1 + 0.033 * \cos \left(\frac{360 n_d}{365} \right) \right] + \left(\cos \varphi \cos \delta \sin \omega_s + \frac{\pi \omega_s}{180} \sin \varphi \sin \delta \right) \quad (8)$$

Where n_d is the day in the year; G_{sc}, solar constant (G_{sc} = 1367 W/m²); φ, latitude of the location (4.08°); δ, the declination angle (°) given by equation (9) and ω_s is the sunset hour angle (°) given by equation (10).

$$\delta = 23.45 \sin \left(360 \frac{248 + n_d}{365} \right) \quad (9)$$

$$\omega_s = \cos^{-1}(-\tan \varphi \tan \delta) \quad (10)$$

The day length, N, is the number of hours between sunrise and sunset given by equation (11).

$$N = \frac{2}{15} \omega_s \quad (11)$$

ii. Assessment of wind energy potential

The monthly average wind speed data was obtained from IGEUBU for (2015 – 2018). This was collected by a station located in the Nyanza-Lac area considered by IGEUBU as having the same climatic conditions as Kigwena (Table 3). NASA data was also used for comparison purposes. Equation (12) was used to extrapolate the collected data (measured at 10 m) to the hub centre of the selected wind turbine [16].

$$\frac{V(Z_1)}{V(Z_2)} = \left[\frac{Z_1}{Z_2} \right]^\alpha \quad (12)$$

Z is height, V (Z), wind speed at the height Z. α is a function of the factors (terrain, wind, speed temperature, surface roughness, time of day and season) and was taken 1/7 [16].

iii. Assessment biomass resources potential

In this study area, the biomass was mainly composed

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of crops residues which could not be sustainable for daily electricity generation [17]. Therefore, it was excluded in the different options.

II.3. Optimization in HOMER software

The available RE resources were evaluated based on 2 major economic output metrics: Net Present Cost (NPC) and Levelized Cost of Energy (LCOE).

$$C_{NPC} = \frac{C_{ann, tot}}{CRF(i, R_{proj})} \quad (13)$$

The capital recovery factor CRF is given by:

$$CRF(i, n) = \frac{(1+i)^n}{(1+i)^n - 1} \quad (14)$$

Where $C_{ann, tot}$ is the total annualized cost, i is the discount rate and n , the number of years (lifetime of the project). The discount rate in Burundi was averaged at 10.18 % (from 2007 to 2018) [18] and averaged inflation rate (from 2014 to 2017) amounted about 16.64 % [19].

$$COE = \frac{C_{ann, tot}}{E_{prim} E_{def} E_{grid, sales}} \quad (15)$$

Where E_{prim} and E_{def} are the total amounts of primary and deferrable load, respectively and $E_{grid, sales}$ is the amount of energy sold to the grid per year.

II.4. Sensitivity analysis

The inflation rate was considered as a sensitivity parameter. The values entered for that variable are: 16.64 %, 18 %, 20 % and 23 %. The investment is not profitable when NPV is negative [20].

III. EXPERIMENTATION

III.1. Electrical parameters of the Generator

The operating generator was a three-phase synchronous of type MJT 250 MA4 B34 with electrical parameters in Table 1. It is a self-exciting, by means of a brushless type excitation system.

Table 1 Electrical parameters of operating alternator

Parameter	Unit	Description
S = 100	kVA	Apparent power
U = 400	V	3 phase voltage system
f = 50	Hz	Frequency
Cos φ = 0.8	-	Power factor
I = 2.2	A	Excitation current
V = 18	V	Excitation voltage

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III.2. Stream flow assessment

The streamflow for only 2 months was evaluated by ABER (Table 2). This was not enough to predict the MHP energy potential. Therefore, the timely recorded data when the plant is in operation mode was used; the values of P and Q recorded every 2 hours throughout the year 2018 were used after calculating their daily and monthly mean values. Estimated water discharge at the entrance of the turbine was found by equations (1), (2) and (3).

Table 2 River flowrates data from ABER

Disch	Value	Year	Description
D1	85 l/s	Feb-2015	Monthly average
D2	32 l/s	Sep-2016	Monthly average
D _{nom}	65 l/s	Mar-2017	Rated discharge

III.3. Options considered for the hybrid micro-grid model

Fig. 2 shows different options considered for the optimization of the MHP.

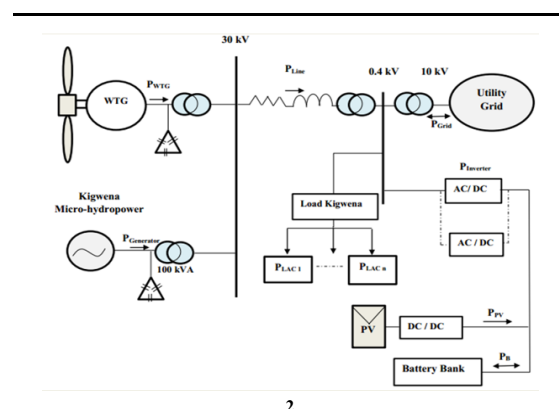


Fig. 2 Options for the HPS model

IV. RESULTS AND DISCUSSION

IV.1. Performance evaluation of Kigwena MHP

Fig. 3 presents the pattern of the generator outputs. This could relate the fluctuations in the power supply. Ossberger cross-flow turbine developed for small hydropower is made for heads between 2.5 and 200 m and can operate with very low river flow rates [21]. A minimum flow ratio of 7.5 % was estimated [22]. Table 3 shows the estimated water flow rates considering a constant turbine efficiency of 75%, generator efficiency of 97 % and hydraulic

efficiency of 85 %. The impact of flow rates on the turbine efficiency was neglected [22].

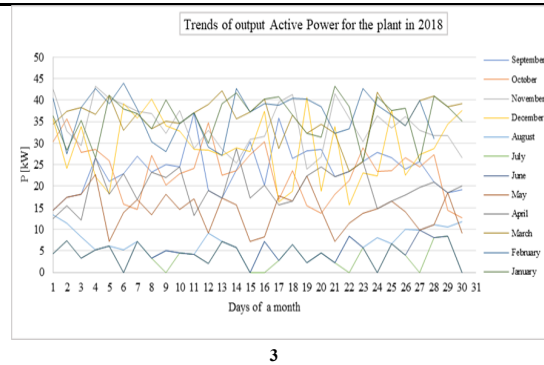


Fig. 3 Generator output: Active power

A descriptive statistical analysis of the power was conducted to check the status of power output of the plant (year 2018). A standard deviation of 17.24 kVA and a mean of 30.66 kVA reflected a high fluctuation in supply. With a peak load of 190.44 kW, there was a need of 159.75 kW to satisfy the load, considering the average power output of the MHP.

IV.2. Alternative energy resources

Using equations (7 to 11), the estimated insolation of the study area is given in Table 4. As there was no specific measured data on the area, NASA data was used as the worst-case scenario. Table 5

recapitulates the wind speed obtained at the heights of 10 m and 50 m using equation (12). The data from IGEPU was used as it was a directly measured data.

IV.3. Optimization results

Fig. 4 shows the results of a best categorized optimum system configurations. A system of 31.1 kW MHP, 74.6 kW solar PV with 50.9 kW inverter and 999,999 kW from the utility grid to smooth the system was found the cheapest configuration option.

Table 3 Estimated water discharge D: year 2018

Year 2018	P [kW]	Q [kVAr]	S [kVA]	P _h [kW]	D [l/s]
Jan	36.05	31.85	48.10	40.50	50.75
Feb	36.49	26.55	45.13	38.00	47.62
Mar	35.88	35.88	50.74	42.73	53.54
Apr	19.65	21.45	29.09	24.49	30.69
May	14.28	16.85	22.08	18.60	23.30
Jun	04.74	02.52	05.37	04.52	05.66
Jul	03.72	02.03	04.24	03.57	04.47
Aug	06.78	03.19	07.49	06.31	07.90
Sep	24.05	23.38	33.54	28.24	35.39
Oct	23.40	24.57	33.93	28.57	35.80
Nov	33.92	32.99	47.32	39.85	49.93
Dec	29.09	28.81	40.94	34.47	43.19

Table 4 Estimated insolation data for the study area

Mid-month	n _d	δ (°)	ω _s (°)	N (hours)	n	H ₀ (kWh/ m ² /day)	n/N	H(kW/ m ² /day)	NASA(kW/ m ² /day)
15-Jan	15	-23.05	88.26	11.77	9.90	9.90	0.84	7.22	4.57
14-Feb	45	-22.17	88.34	11.78	10.10	9.97	0.86	7.35	4.94
15-Mar	74	-15.82	88.84	11.85	8.40	10.36	0.71	6.85	4.99
15-Apr	105	-4.81	89.66	11.95	8.00	10.73	0.67	6.88	4.92
15-May	135	7.15	90.51	12.07	8.10	10.68	0.67	6.86	4.90
15-Jun	166	17.52	91.29	12.17	7.30	10.26	0.60	6.22	5.05
15-Jul	196	22.93	91.73	12.23	5.80	9.91	0.47	5.37	5.27
15-Aug	227	22.24	91.67	12.22	4.90	9.96	0.40	5.02	5.38
15-Sep	258	15.36	91.12	12.15	7.90	10.38	0.65	6.56	5.25
15-Oct	288	4.61	90.33	12.04	9.10	10.73	0.76	7.35	4.72
15-Nov	319	-7.72	89.45	11.93	9.50	10.67	0.80	7.53	4.41
15-Dec	349	-17.65	88.70	11.83	9.70	10.26	0.82	7.37	4.49
Average								6.72	4.91

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Table 5 Estimated insolation data for the study area

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec	Average
NASA	(50 m)	2.48	2.53	2.49	2.97	3.9	4.29	4.25	4.09	3.44	2.75	2.42	2.17	3.15
IGEBU	(10 m)	0.23	0.29	0.29	0.26	0.41	0.64	0.75	0.85	0.9	0.85	0.6	0.38	0.54
	(50 m)	0.29	0.36	0.36	0.32	0.52	0.8	0.94	1.07	1.13	1.07	0.75	0.47	0.68

This option reflects a COE of USD 0.0334/kWh and NPC of USD 732,432. The LCOE was found lower than the existing tariff in Burundi, as of 2019. This was explained by the fact that there was excess of electricity injected in the utility grid and that the grid power price and grid sellback price were considered USD 0.071/kWh and USD 0.030/kWh respectively. According to the Utility of Water and Electricity Distribution in Burundi (REGIDESO), the tariffs for electricity, as of 2019, ranged from 41 BuF/kWh for low energy consumers to 127 BuF/kWh for consumption over 750 kWh per 2 months. The ABER tariffs also vary from 56 BuF/kWh to 72 BuF/kWh (USD 1 = 1,850 BuF as of 2019).

Optimization Results														
Left Double Click on a particular system to see its detailed Simulation Results.														
Export	Architecture	Grid	Hyd100	Converter	Dispatch	COE	NPC	Operating cost	Initial capital	Ren. Fac.	System			
		(kW)	(kW)	(kW)	(kW)	(\$/kWh)	(\$/kWh)	(\$/kWh)	(\$/kWh)	(%)				
	74.6	999.999	31.1	50.9	CC	\$0.0333	\$732,432	\$9,471	\$196,221	71.9				
	73.1	1	999.999	31.1	50.3	CC	\$0.0335	\$734,174	\$9,487	\$193,842	71.7			
	74.4	1	999.999	31.1	50.2	CC	\$0.0346	\$760,850	\$9,604	\$193,667	71.9			
	76.8	1	999.999	31.1	51.8	CC	\$0.0344	\$762,399	\$9,548	\$218,636	72.2			
			999.999	31.1		CC	\$0.0425	\$808,764	\$13,377	\$46,650	53.2			
			999.999	31.1	0.161	CC	\$0.0426	\$810,573	\$13,402	\$46,698	53.2			
			999.999	31.1		CC	\$0.0440	\$837,176	\$13,359	\$64,650	53.2			
			999.999	31.1	2.48	CC	\$0.0442	\$842,068	\$13,612	\$65,995	53.2			

Fig. 4 Cost summary for feasible options

IV.4. Energy generation

Fig. 5 describes the monthly average energy production of the optimal hybrid system composed of utility grid, the MHP and solar PV system. The expected annual energy production from the system is 393,500 kWh, of which almost half (45.2 %) is contributed by the MHP. The utility grid has a significant contribution in this configuration; energy excess is directly fed into the grid (grid sales) and taken back to the community (grid purchases) when there is deficiency from the system.

i. Excess electricity and unmet electricity load

Table 6 presents the yearly electricity excess. The system meets the load along the whole year with excess of 1,837 kWh/year without capacity shortage.

ii. Emissions analysis

Table 7 displays the emissions analysis.

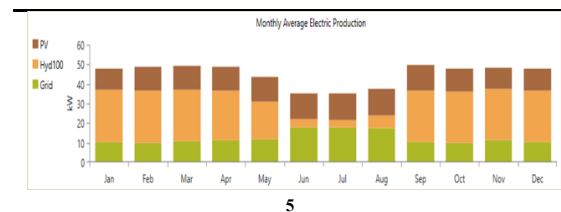


Fig. 5 Monthly energy production

Table 6 Annual total of excess electricity

Quantity	kWh/yr	%
Excess Electricity	1,837	0.467
Unmet Electric Load	0	0
Capacity Shortage	0	0

Table 7 Emissions analysis for the optimum HPS

Quantity	Value	Units
Carbon Dioxide	-856	kg/yr
Sulfur Dioxide	-3.71	kg/yr
Nitrogen Oxides	-1.81	kg/yr

In 25 years (lifetime), the system would reduce (conservation equivalent) 856 kg/year carbon dioxide emission, 3.71 kg/year sulphur dioxide emission and 1.81 kg/year nitrogen oxides emission.

IV.5. Sensitivity analysis

Fig. 6 shows that the inflation rate beyond 20 % would make the project economically not viable.

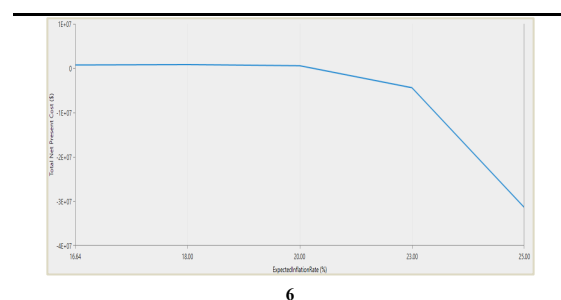


Fig. 6 Total NPV with expected inflation rate

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IV. CONCLUSIONS

A reliable and cost-effective hybrid power system to upgrade the power supply of Kigwena MHP was studied. In overall optimization results, different configurations with their total NPC and LCOE obtained, the most optimum combination was solar PV, MHP and utility grid with the lowest LCOE of US\$ 0.0334 /kWh. The suggested system responds to the persistent energy shortage for the village while promoting clean energy. This solution is subjected to change the lifestyle of rural communities in Burundi. This work concludes that Burundi needs to invest in RE such as MHP and solar PV in order to meet the growing energy demand. This is because, it has a huge untapped potential in hydropower and solar. Furthermore, as RE are intermittent sources of energy, their combinations in hybrid system would help to overcome the weakness of one to another and would offer a possibility for higher reliability in power supply. Both on/off-grid system should be encouraged to decentralize the power supply.

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